

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:

Exploring Processes, Threads, Handles, and Windows Registry

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Introduction

This lab explores Windows operating system internals using the Sysinternals Suite, specifically Process Explorer, and the Windows Registry Editor. The lab demonstrates how processes, threads, and handles function, and how registry settings affect application behavior.

Objectives

- Explore active processes using Process Explorer
- Identify parent and child processes
- Examine threads and handles
- Modify and observe changes in the Windows Registry

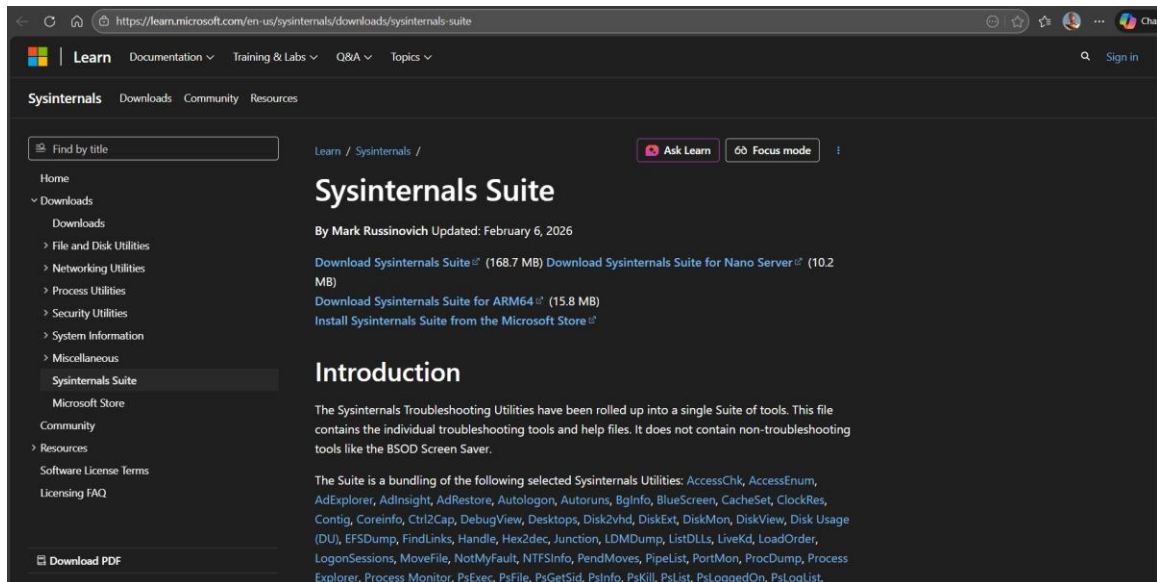
Required Resources

- Windows PC with internet access
- Sysinternals Suite
- Process Explorer (procexp.exe)
- Registry Editor (regedit)

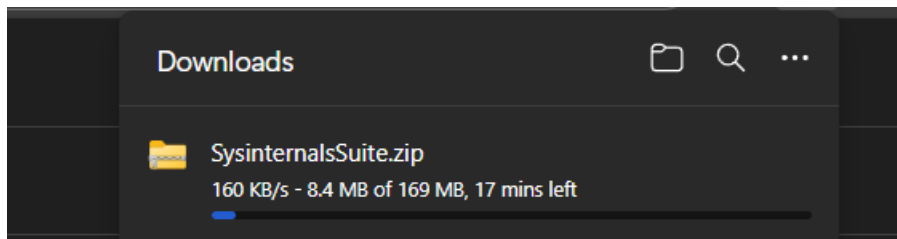
Part 1: Exploring Processes

Step 1: Download Windows Sysinternals Suite

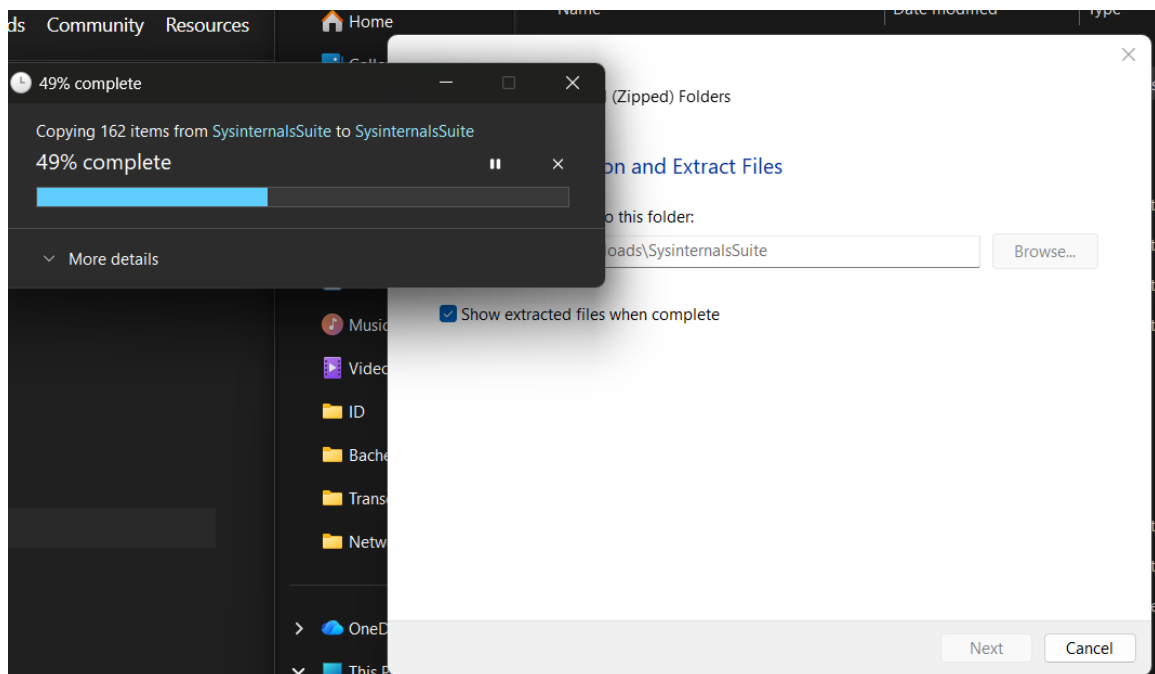
1. Navigate to the Sysinternals download page.



2. Download the Windows Sysinternals Suite.



3. Extract the downloaded folder.

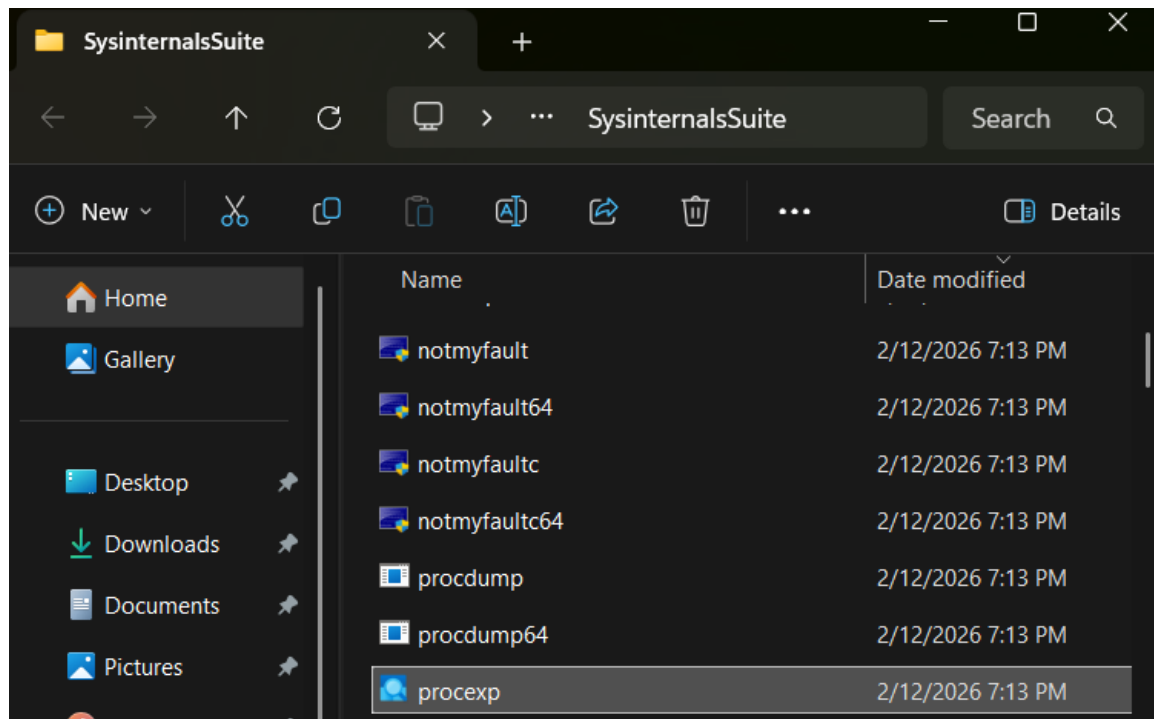


4. Keep the browser open for later steps.

Step 2: Explore an Active Process

1. Open the extracted Sysinternals folder.

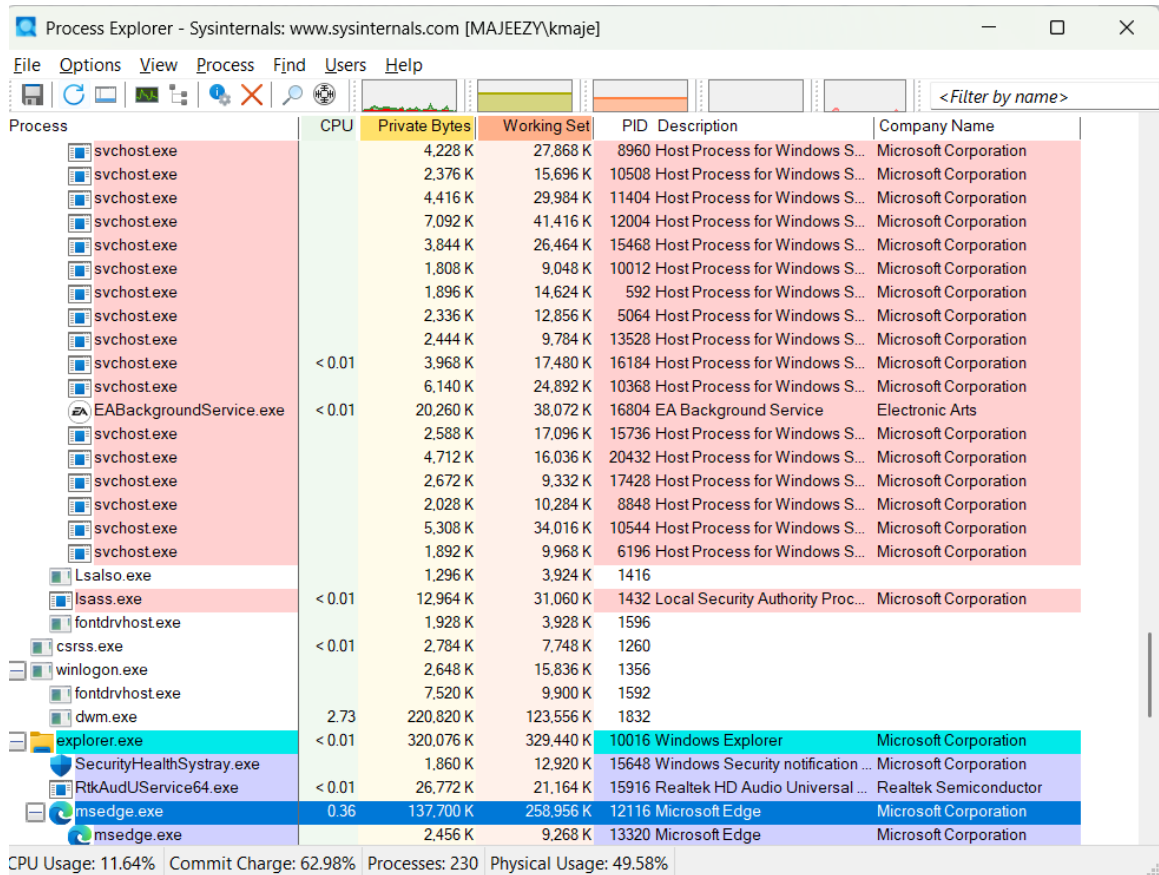
2. Launch procexp.exe.



3. Accept the License Agreement.

4. Use the "Find Window's Process" tool and drag it onto the web browser window.

5. Observe the highlighted browser process.



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
svchost.exe		4,228 K	27,868 K	8960	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,376 K	15,696 K	10508	Host Process for Windows S...	Microsoft Corporation
svchost.exe		4,416 K	29,984 K	11404	Host Process for Windows S...	Microsoft Corporation
svchost.exe		7,092 K	41,416 K	12004	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,844 K	26,464 K	15468	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,808 K	9,048 K	10012	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,896 K	14,624 K	592	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,336 K	12,856 K	5064	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,444 K	9,784 K	13528	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	3,968 K	17,480 K	16184	Host Process for Windows S...	Microsoft Corporation
svchost.exe		6,140 K	24,892 K	10368	Host Process for Windows S...	Microsoft Corporation
EABackgroundService.exe	< 0.01	20,260 K	38,072 K	16804	EA Background Service	Electronic Arts
svchost.exe		2,588 K	17,096 K	15736	Host Process for Windows S...	Microsoft Corporation
svchost.exe		4,712 K	16,036 K	20432	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,672 K	9,332 K	17428	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,028 K	10,284 K	8848	Host Process for Windows S...	Microsoft Corporation
svchost.exe		5,308 K	34,016 K	10544	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,892 K	9,968 K	6196	Host Process for Windows S...	Microsoft Corporation
lsalss.exe		1,296 K	3,924 K	1416		
lsass.exe	< 0.01	12,964 K	31,060 K	1432	Local Security Authority Proc...	Microsoft Corporation
fontdrvhost.exe		1,928 K	3,928 K	1596		
csrss.exe	< 0.01	2,784 K	7,748 K	1260		
winlogon.exe		2,648 K	15,836 K	1356		
fontdrvhost.exe		7,520 K	9,900 K	1592		
dwm.exe	2.73	220,820 K	123,556 K	1832		
explorer.exe	< 0.01	320,076 K	329,440 K	10016	Windows Explorer	Microsoft Corporation
SecurityHealthSystray.exe		1,860 K	12,920 K	15648	Windows Security notification ...	Microsoft Corporation
RtkAudUService64.exe	< 0.01	26,772 K	21,164 K	15916	Realtek HD Audio Universal ...	Realtek Semiconductor
msedge.exe	0.36	137,700 K	258,956 K	12116	Microsoft Edge	Microsoft Corporation
msedge.exe		2,456 K	9,268 K	13320	Microsoft Edge	Microsoft Corporation

CPU Usage: 11.64% Commit Charge: 62.98% Processes: 230 Physical Usage: 49.58%

6. Right-click the browser process and select "Kill Process".

Process Explorer - Sysinternals: www.sysinternals.com [MAJEEZY\kmaje]

File Options View Process Find Users Help

<Filter by name>

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
svchost.exe		9,712 K	24,168 K	25112	Host Process for Windows S...	Microsoft Corporation
msiexec.exe		7,872 K	20,956 K	11272	Windows® installer	Microsoft Corporation
svchost.exe	0.72	2,792 K	18,756 K	20704	Host Process for Windows S...	Microsoft Corporation
lsalco.exe		1,296 K	3,916 K	1416		
lsass.exe	< 0.01	13,176 K	31,548 K	1432	Local Security Authority Proc...	Microsoft Corporation
fontdrvhost.exe		1,928 K	3,928 K	1596		
csrss.exe	0.18	2,848 K	7,780 K	1260		
winlogon.exe		2,796 K	16,436 K	1356		
fontdrvhost.exe		7,520 K	8,240 K	1592		
dmw.exe				1832		
explorer.exe				K	10016 Windows Explorer	Microsoft Corporation
SecurityHealthSystray.exe				K	15648 Windows Security notification ...	Microsoft Corporation
RtkAudUService64.exe				K	15916 Realtek HD Audio Universal ...	Realtek Semiconductor
msedge.exe				K	12116 Microsoft Edge	Microsoft Corporation
msedge.exe				K	13320 Microsoft Edge	Microsoft Corporation
msedge.exe				K	13452 Microsoft Edge	Microsoft Corporation
msedge.exe				K	3348 Microsoft Edge	Microsoft Corporation
msedge.exe				K	8768 Microsoft Edge	Microsoft Corporation
msedge.exe				K	14724 Microsoft Edge	Microsoft Corporation
msedge.exe				K	16724 Microsoft Edge	Microsoft Corporation
msedge.exe				K	16436 Microsoft Edge	Microsoft Corporation
msedge.exe				K	15740 Microsoft Edge	Microsoft Corporation
msedge.exe				K	17944 Microsoft Edge	Microsoft Corporation
WINWORD.EXE				K	9108 Microsoft Word	Microsoft Corporation
msedge.exe				K	9648 Microsoft Edge	Microsoft Corporation
msedge.exe				K	25088 Microsoft Edge	Microsoft Corporation
OneDrive.exe				K	9612 Microsoft OneDrive	Microsoft Corporation
SnippingTool.exe				K	6544	
procexp.exe				K	4524 Sysinternals Process Explorer	Sysinternals - www.sysinter...
procexp64.exe	0.72	37,436 K	57,360 K	K	17692 Sysinternals Process Explorer	Sysinternals - www.sysinter...

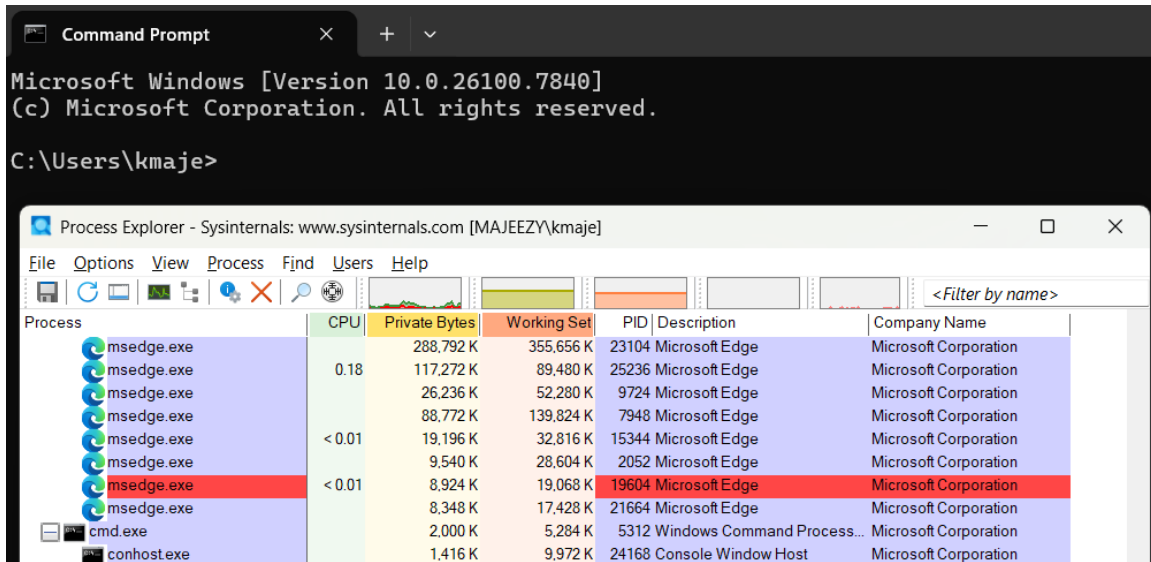
CPU Usage: 24.84% Commit Charge: 65.44% Processes: 245 Physical Usage: 53.89%

Observation:

When the browser process was killed, the browser window immediately closed. Terminating the process stops the application execution completely.

Step 3: Start Another Process

1. Open Command Prompt (cmd.exe).
2. Use the Find Window's Process tool to locate cmd.exe in Process Explorer.



3. Observe that explorer.exe is the parent process.
4. Observe that conhost.exe is the child process.
5. In Command Prompt, run a ping command (e.g., ping -t www.cisco.com).

```
Command Prompt
Microsoft Windows [Version 10.0.26100.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kmaje>ping www.cisco.com

Pinging e2867.dsca.akamaiedge.net [2.22.192.103] with 32 bytes of data:
Reply from 2.22.192.103: bytes=32 time=59ms TTL=50
Reply from 2.22.192.103: bytes=32 time=38ms TTL=50
Reply from 2.22.192.103: bytes=32 time=37ms TTL=50
Reply from 2.22.192.103: bytes=32 time=33ms TTL=50

Ping statistics for 2.22.192.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 33ms, Maximum = 59ms, Average = 41ms

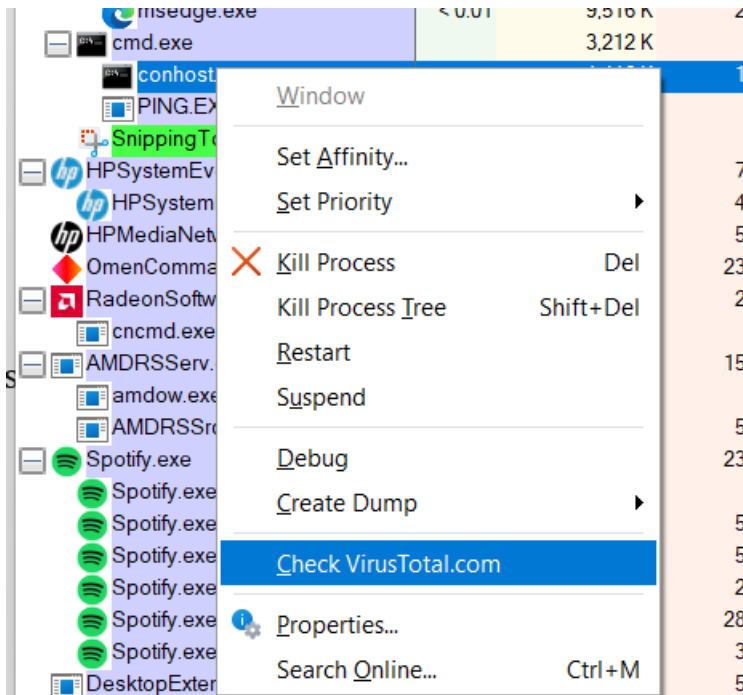
C:\Users\kmaje>
```

cmd.exe	Advanced Micro Devices, Inc.	2,180 K	5,872 K	5312 Windows Command Process...	Microsoft Corporation
conhost.exe		1,416 K	10,032 K	24168 Console Window Host	Microsoft Corporation
PING.EXE		<0.01	1,112 K	21956 TCP/IP Ping Command	Microsoft Corporation

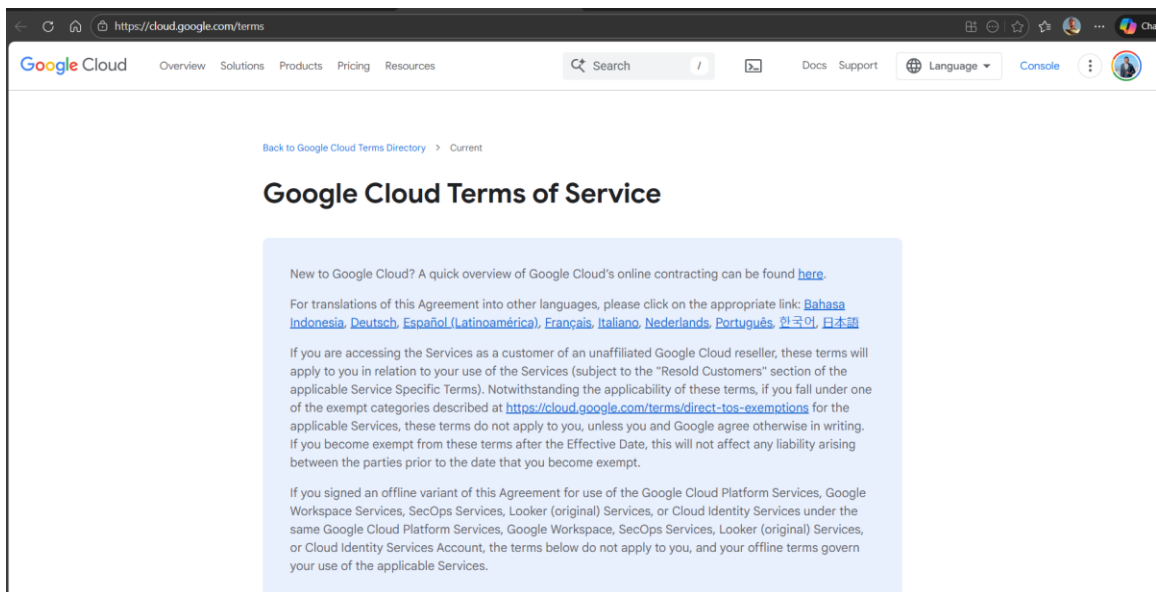
Observation:

During the ping process, CPU activity increased and cmd.exe showed active resource usage. The process continuously sent and received packets.

6. Right-click conhost.exe and select "Check VirusTotal".



7. Review results in browser.



8. Kill cmd.exe process.

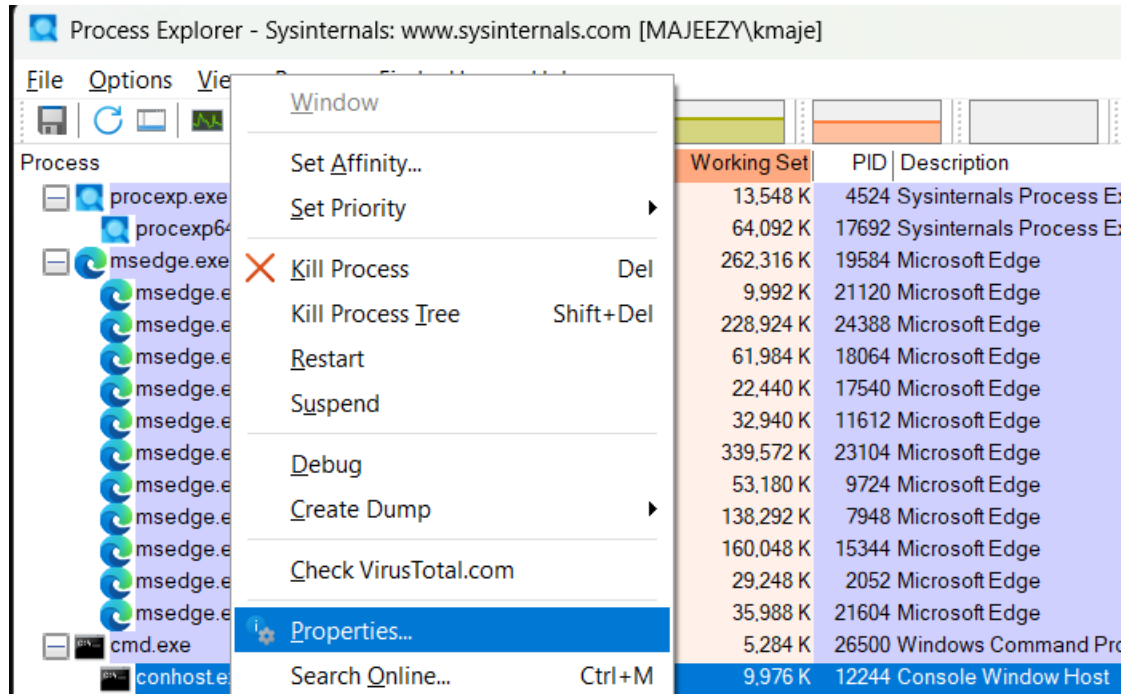
Observation:

When cmd.exe was terminated, the child process conhost.exe also closed automatically because it depends on the parent process.

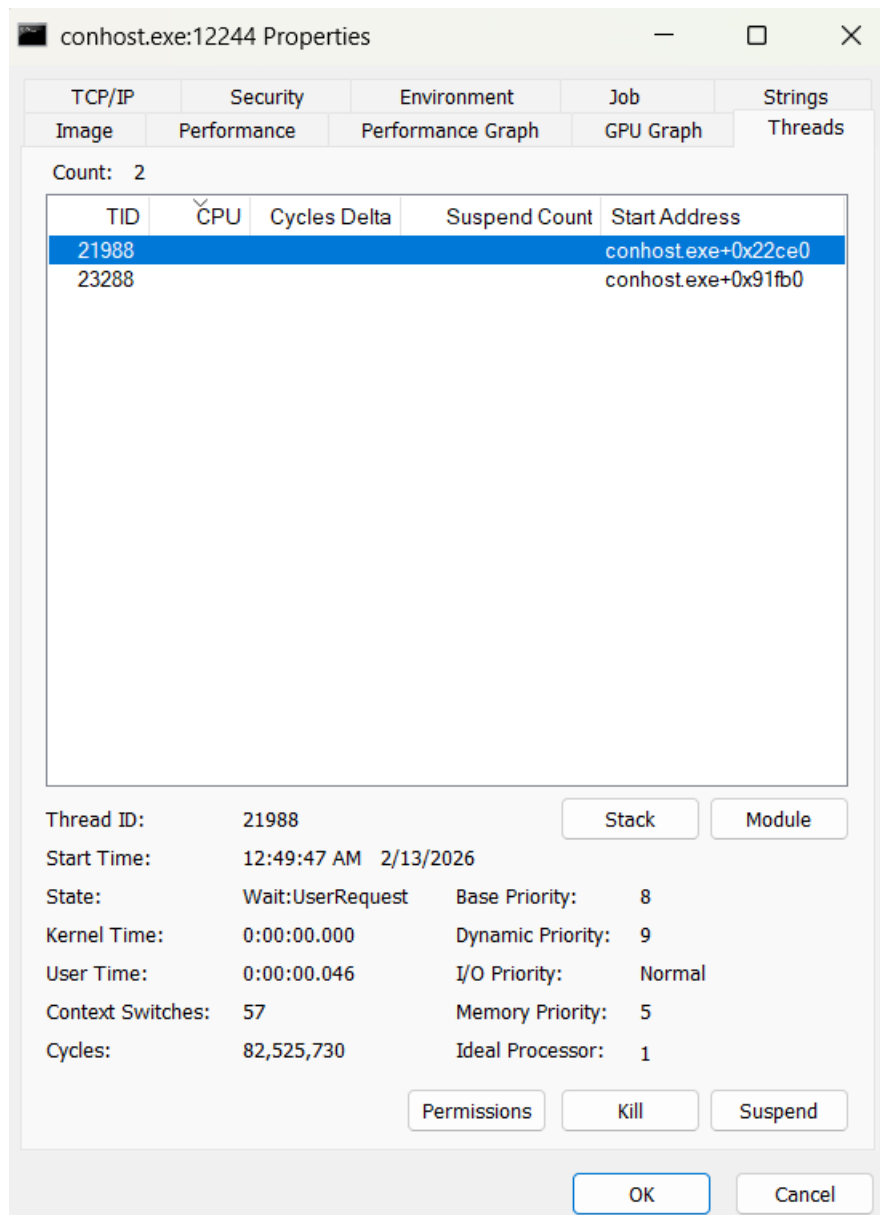
Part 2: Exploring Threads and Handles

Step 1: Explore Threads

1. Open Command Prompt again.
2. In Process Explorer, right-click conhost.exe and select Properties.



4. Click the Threads tab.

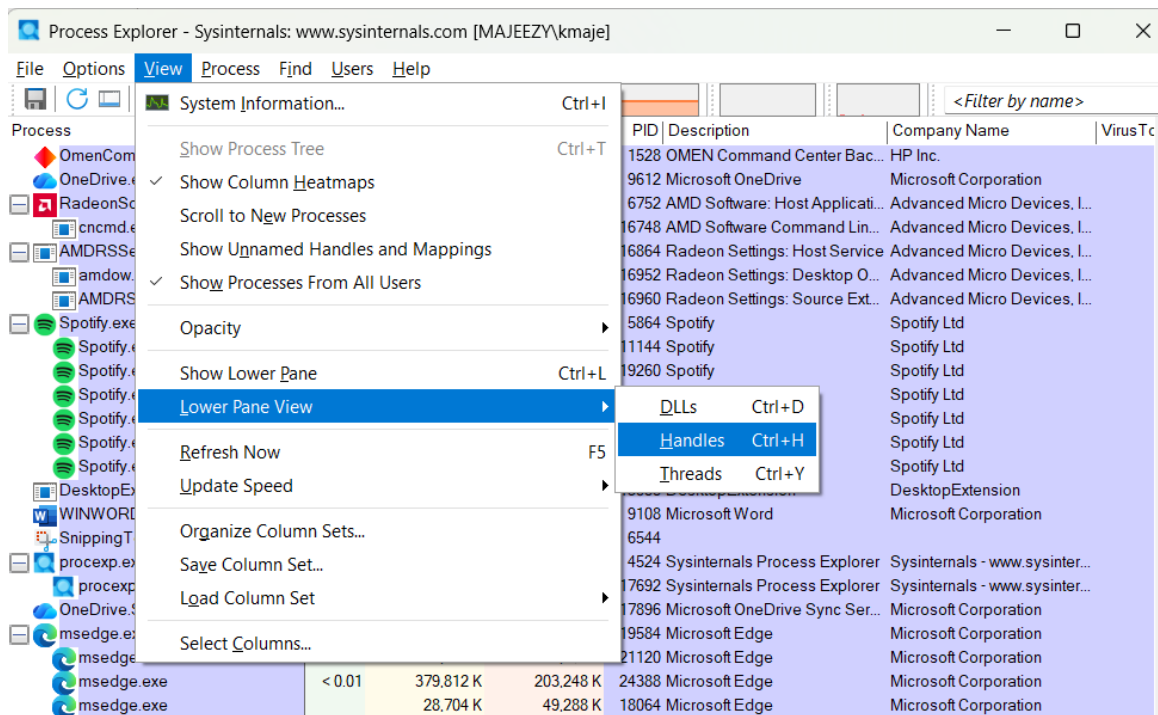


Observation:

The Threads tab displays Thread ID, CPU usage, state, start address, priority, and stack details. Threads represent units of execution within a process.

Step 2: Explore Handles

1. In Process Explorer, click View > Lower Pane View > Handles.



2. Examine handles associated with conhost.exe.

Handles		DLLs	Threads
Type	Name		
Key	HKCU\Software\Classes		
Key	HKCU\Software\Classes\PackagedCom		
Key	HKCU\Software\Classes\PackagedCom\Package		
Key	HKCR\PackagedCom\Package		
Key	HKLM		
Key	HKCU\Software\Classes		
Key	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Diagnostics\DiagTrack		
Key	HKCR\PackagedCom\InterfaceIndex		
Key	HKCU\Software\Classes		
Mutant	\Sessions\1\BaseNamedObjects\SM0:12244:304:WilStaging_02		
Mutant	\Sessions\1\BaseNamedObjects\SM0:12244:120:WilError_03		
Section	\BaseNamedObjects__ComCatalogCache__		
Section	\BaseNamedObjects__ComCatalogCache__		
Semaphore	\Sessions\1\BaseNamedObjects\SM0:12244:304:WilStaging_02_p0		
Semaphore	\Sessions\1\BaseNamedObjects\SM0:12244:304:WilStaging_02_p0h		
Semaphore	\Sessions\1\BaseNamedObjects\SM0:12244:120:WilError_03_p0		
Semaphore	\Sessions\1\BaseNamedObjects\SM0:12244:120:WilError_03_p0h		
Thread	conhost.exe(12244): 23288		
Thread	conhost.exe(12244): 21988		
Thread	conhost.exe(12244): 21988		
WindowStation	\Sessions\1\Windows\WindowStations\WinSta0		
WindowStation	\Sessions\1\Windows\WindowStations\WinSta0		
CPU Usage: 19.66%		Commit Charge: 64.45%	Processes: 245
		Physical Usage: 52.17%	

Observation:

Handles pointed to system resources such as files, registry keys, memory sections, events, and console input/output objects.

Conclusion

This lab provided practical experience analyzing Windows processes, threads, and handles. It also demonstrated how registry settings affect application behavior. The exercises reinforced understanding of process management and system configuration in Windows.